

Coarse mismatch vs. factor-level diagnosis

$$\text{LEMON}(e, T) = \sum_i w_i \cdot m_i$$

w_i = factor weight m_i = evidence match in text (0, 0.5, 1)

✓ preserved ½ partial × missing/contradicted

A. Polarity error

Triple: Aspirin — inhibits — COX-1

Text: "Aspirin activates COX-1."

MINE-style

nodes: Aspirin ✓, COX-1 ✓
edge/fact: inhibits ×
score ≈ 0.50

LEMON-Factor

factor	weight×match	contrib.
chemical actor	0.20 × 1.0	0.20
protein target	0.20 × 1.0	0.20
regulation relation	0.15 × 0.5	0.075
negative polarity	0.30 × 0.0	0.000
direction chem→protein	0.15 × 1.0	0.150

LEMON score = 0.625

0.20	0.20	0.075	0.000	0.150
chemical actor	protein target	regulation relation	negative polarity	direction chem → protein

Diagnosis: polarity contradicted

B. Relation-class loss

Triple: Alan Bean — birthPlace — Wheeler, Texas

Text: "Alan Bean lived in Wheeler, Texas."

MINE-style

nodes: Alan Bean ✓
Wheeler ✓
edge/fact: birthPlace ×
score ≈ 0.50

LEMON-Factor

factor	weight×match	contrib.
person subject	0.25 × 1.0	0.25
place object	0.25 × 1.0	0.25
birth relation	0.35 × 0.0	0.00
direction person→place	0.15 × 1.0	0.15

LEMON score = 0.65

0.25	0.25	0.00	0.15
person subject	place object	birth relation	direction person → place

Diagnosis: birth relation missing

Same coarse failure, different semantic diagnosis: MINE-style matching collapses both cases into relation mismatch, while LEMON-Factor preserves graded partial credit and localizes the error.